AI FOR INDUSTRIES: Using Intelligence In Real Time

If AI is to be used in industrial applications, parallel computing with the shortest response times in real time is often required. This places new demands on embedded computer technology, such as computer-on-modules (COMs)

Zeljko Loncaric is marketing engineer at congatec AG

Fig. 1: Migration to AI is a simple task with COMs, as existing COM designs can easily be replaced with other processor technologies (Credit: congatec)

The artificial intelligence (AI) market is currently experiencing an incredible boom. According to ResearchAndMarkets, the global AI market is expected to grow by 36.6 per cent annually to US$ 190.61 billion by 2025. AI is already all around us today. In our personal lives we experience AI, for example, in product, film and music recommendations as well as service hotlines, where AI bots answer general customer questions completely autonomously.

AI is also behind smartphone features, such as face and gesture recognition and personal (voice) assistants. However, according to Deloitte's study "Exponential Technologies in Manufacturing," one of the most important drivers for new AI is also the production sector. Next to less-critical use cases—such as AI for predictive maintenance or demand forecasting—industrial AI can also be used in areas of application that require real-time capabilities. These include, for example:

- Industrial image processing, where AI helps capture many different states and/or product features, and evaluates these via intelligent pattern recognition to enable more reliable quality control. Forbes states that AI can increase error detection rate by up to 90 per cent.
- Cooperative and collaborative robots that share the same work space with humans can react flexibly to unforeseen events, making decisions based on situational awareness.
- Similar requirements apply to autonomous industrial vehicles.
- AI applications in the semiconductor industry showed, for example, 30 per cent reduction in reject rates when using Big Data to analyse root-cause data, which must be collected with high precision and in real time.
- Lastly, there is the large area of real-time production planning and control of Industry 4.0 factories. Here, AI helps optimise the processes at the edge, thereby increasing utilisation of machines and, ultimately, productivity of the entire factory.

Use of AI in industrial environments requires highly-advanced integrated logic. This is because, in fast processes—as in the case of inspection systems—there is often no further control authority. This is different, for example, in medical technology, where AI results of automatic